

Analytics Engineer (w/m/d) - Coding Exercise

Imagine you are working as an Analytics Engineer for a company offering a mobile app. This app collects millions of data points from users every day. Your goal is to transform and organize this data so that not only you can derive valuable insights from it, but also product managers and marketers can easily query it.

Attached to this exercise you can find an SQLite database containing two tables. In the following, we give a brief description of the data.

This first table (`events`) contains 43,479 telemetric events for one week from a hypothetical mobile app. All users contained in the database are new users. Each row in this table describes a single event by a user and contains the following columns:

- `user_id`
- `event_name`
- `event_timestamp` (unix timestamp measured in microseconds)
- `platform`
- `os` (= operating system)
- `country`
- `ad_revenue` (only set for particular events related to monetization)
- `tracker_name` (the name of the campaign if a user was acquired via paid marketing or "Unattributed" if this is an organic user.

The second table (`user_acquisition`) contains information about user acquisition campaigns. These campaigns were run to acquire new users via different ad networks. The data contains for each day and campaign a tracker name (i.e., the name of the campaign) and the amount spent on this day for this campaign. To be more precise, the table contains the following columns:

- `date`
- `tracker_name`
- `costs`

Please answer the following questions and implement the necessary tasks in a programming language of your choice (**hint:** the Python standard library offers a module `sqlite3` which can be easily used for the task to read a file based

database. Of course, you are also free to choose other packages or programming languages).

1. What's the total number of users present in the dataset?
2. List the number of installs per country.
3. In this exercise, you will calculate the retention for a specific cohort.
 - How many users installed the app on August 2, 2022 in Germany on Android?
 - How many of these users are active on the first, third, and fourteenth day after the install respectively? (I.e., count users for all three days separately)
 - How much are those in percent? These are called day 1, day 3, and day 14 retention.
4. Create a view named **marketing** that provides the following columns per day and per campaign:
 - **day**
 - **tracker_name**
 - **number_of_installs**
 - **costs** (costs spent on this day for the specific campaign)
 - **total_revenue** (revenue from the users acquired on this day from this campaign. Make use of the column **ad_revenue** from **events**)
5. Query the view **marketing** and report the Costs per Install (CPI) on August 6, 2022, for campaign "google_campaign1"?

Final Remarks

- Please submit your documented code along with instructions on how to run the code and the answers to the questions above.
- Please only spend 120 minutes on the exercise. We know that it is challenging to complete all tasks but please respect the time limit.

Attachment

- exercise.db.zip